

User's Manual

Thor 16 Ryzen 9 RTX 5070 Ti

Thor 16 Ryzen 9 RTX 5080

Thor 16 Ryzen 9 RTX 5090

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The illustrations in this user's manual are for reference only.
Actual product specifications may vary with territories.

The information in this user's manual is subject to change without notice.

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Chapter 1

Preface

1.1 Regulations Information

☒ CE compliance

This device is classed as a information technology equipment (ITE) in class B and is intended for use in living room and office. The CE-mark approves the conformity by the EU-guidelines:

- EMC Directive 2014/30/EU,
- Low Voltage Directive 2014/35/EU ,
- RF Directive 2014/53/EU

For RE-Directive 2014/53/EU

All operational modes:

The adapter shall be installed near the equipment and shall be easily accessi-ble.

2.4GHz:802.11b,802.11g,802.11n-20,802.11n-4,Bluetooth

5GHz:802.11a,802.11ac-20,802.11ac-40,802.11ac-80,802.11ac-160

5.8GHz:802.11ax-20,802.11ax-40,802.11ax-80,802.11ax-160

6GHz:802.11ax-20,802.11ax-40,802.11ax-80,802.11ax-160

The frequency and the maximum transmitted power in EU and UK are listed below:

2400-2485MHz:20 dBm

5150-5250MHz:23 dBm

5470-5725MHz:23 dBm

5725-5875MHz:13.95 dBm

5945-6425MHz:23 dBm

This device operates in the 5150-5350MHz and 5925-6425MHz frequency range and is restricted to indoor use only. Outdoor operation in this range is prohibited.

SAR/DAS refers to the rate at which the body absorbs RF energy.

The SAR limit set by the ICNIRP Guidelines is 2.0 W/kg(10g).

Testing for SAR is conducted using standard operating positions accepted by the EN standard.

During testing, the radio is set to its highest transmission levels and placed in positions that simulate use against the body.

This product is compliant with ICNIRP Guidelines with respect to Electromagnetic Fields (EMF) which specifies a Specific Absorption Rate (SAR) limit of 2W/kg. DAS*/SAR: 0.10W/kg (corps/body)



Rechargeable Battery Notice

Do not

1. Throw into fire or a hot oven, or mechanically crush or cutting of a BATTERY
2. Throw or immerse into water
3. Heat to more than 60°C
4. Repaire or disassemble
5. Leave in an extremely low air pressure environment
6. Leave in an extremely high-temperature environment

A power cord is connected to a socket-outlet with earthing connection.



The unit can be operated at an ambient temperature of max. 35°C (95°F). Do not subject it to temperatures below 5°C (41°F) or above 40°C (104 °F).

CAUTION: RISK OF EXPLOSION IF BATTERY IS REPLACED BY AN INCORRECT TYPE DISPOSE OF USED BATTERIES ACCORDING TO THE INSTRUCTIONS.



PREVENTION OF HEARING LOSS



CAUTION: Listening to music at high volume levels and for extended durations can damage one's hearing. In order to reduce the risk of damage to hearing, one should lower the volume to a safe, comfortable level, and reduce the amount of time listening at high levels. Headsets should comply with EN 50332-2 requirements.

FCC Information

FEDERAL COMMUNICATIONS COMMISSION INTERFERENCE STATEMENT

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/ TV technician for help.

RF Exposure Information(SAR)

This device meets the government's requirements for exposure to radio waves. This device is designed and manufactured not to exceed the emission limits for exposure to radio frequency (RF) energy set by the Federal Communications Commission of the U.S. Government.

The exposure standard employs a unit of measurement known as the Specific Absorption Rate, or SAR. The SAR limit set by the FCC is 1.6 W/kg. Tests for SAR are conducted using standard operating positions accepted by the FCC with the EUT transmitting at the specified power level in different channels.

The FCC has granted an Equipment Authorization for this device with all reported SAR levels evaluated as in compliance with the FCC RF exposure guidelines. SAR information on this device is on file with the FCC and can be found under the Display Grant section of www.fcc.gov/eot/ea/fccid after searching on FCC ID:RAS-MT7922A22M.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.



CAUTION: Any changes or modifications not expressly approved by the grantee of this device could void the user's authority to operate the equipment.

1.2 Safety Instructions



- 1** Do not apply heavy pressure to the computer or subject it to any form of strong impact as this can damage the computer's components or otherwise cause it to malfunction.
- 2** Never cover or block the air vents including those located at the base of the computer. Never cover your computer or AC adapter with any object.
- 3** To keep your computer in prime operating condition, protect your work area from direct sunlight.
- 4** Do NOT expose to or use near liquid, rain, or moisture.
Do NOT use the modem during electrical storms.
- 5** Do not use or expose this device around magnetic fields as magnetic interference may affect the performance of the device.

1.3 Conventions for this Manual



CAUTION : This message contains important information that must be followed for safe operation.



NOTE : This message contains information for special situations.

1.4 Release History

Version	Date	Revision Note
1.0	11/2024	Initial release

Chapter 2

Getting to know the basics

2.1 Product Specification

This User's Manual provides technical information of instructions and illustrations on how to operate this notebook for the customer. Please read this manual carefully before using this notebook.

· Physical Characteristic

■ Dimension

356.68 x 245.02 x 25.6mm $\pm 1\%$

356.68 x 245.02 x 27.5mm $\pm 1\%$ (w/ mini LED)

Weight: 2.4Kg +/- 5% (depend upon the actually shipping product)

· CPU

■ Support Processor

AMD Fire Range HX55W

· GPU

■ Support Processor

NVIDIA GN22 Board

· Memory

■ RAM

DDR5 5600MHz

· Display

■ LCD

16", 16:10 WQXGA

2.5K 2560x1600

· I/O Port

DC-in	x 1
USB	USB-A @Left x1 (USB3.2 Gen1, 5Gbps) USB-A @Right x2 (USB3.2 Gen1, 5Gbps) USB-C @Rear x1 (USB3.2 Gen2, 10Gbps+DP1.4+PD140W) support 20V only USB-C @Left x1 (USB3.2 Gen2, 10Gbps+DP1.4+PD100W) support 20V only
Mini DP	x 1
RJ45	x 1
HDMI	x 1 V2.1 (dGPU, FRL-12G, 48Gbps)
3.5mm UAJ Jack	x 1
Card reader	x 1 SD/SDHC/SDXC, SD7.0

CAUTION: To prevent system overload, Type-C is prohibited from being under load when using a 100W PD charger

· Audio

■ Codec

Smart Amplifier
Realtek ALC1308 / ALC274
Nahimic

■ Speaker/MIC

Build-in 2 speakers
Speaker : 2W/each

· Communication Port

■ Wireless LAN +BT

M.2 2230 with PCIe interface

(Wi-Fi 6E MT7922A22M/RZ616/AW-XB591NF)

IEEE802.11 a/b/g/n/ac/ax support

(Wi-Fi 7 MT7925B22M/ RZ717/AW-EB600 reserved)

IEEE802.11 a/b/g/n/ac/ax/be support

■ LAN

Supports 10/100/1000/2500 Mb/sec

· Input

■ Keyboard

Membrane with Full-Num

travel = 1.4 ± 0.2 mm

Backlit = Per Key RGB

N-key rollover+Anti-Ghost

■ Pointing Device

TDA2174: 121.2 x 74.9 mm

Glass-like

Glass(option)

Support palm rejection

· Webcam

- Infrared capability for facial recognition camera (Option)

HD+IR Hybrid Camera D-MIC x2, 4mm

FHD & IR dual-sensor D-MIC x2, 4mm (Option)

· Power

- AC Adapter

Automatics Voltage adjustment between 100 and 240VAC 50/60Hz,

420 Watts (20V/21A) · 3 Pins (X6FR558Y/X6FR559Y)

330 Watts (20V/16.5A) · 3 Pins (X6FR57TY)

280 Watts (20V/14A) · 3 Pins (X6FR57TY w/mini LED Panel)

250 Watts (20V/12.5A) · 3 Pins (X6FR57TY)

- Battery (By model)

Li- polymer Battery, SW Gas Gauge IC, soft pack,

4S1P 5200mAh 80Wh

4S1P 6450mAh 99Wh (Option)



CAUTION: MODEL IS DESIGNED TO USE WITH THE

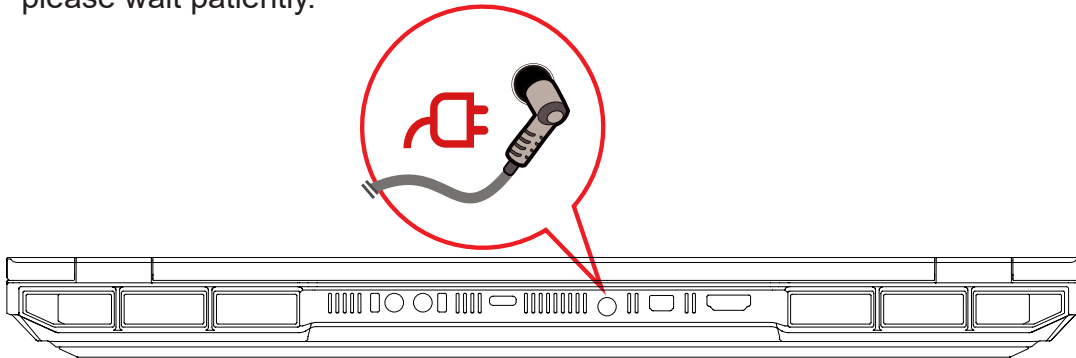
DC INPUT: 420 Watts (20V/21A) / 330 Watts (20V/16.5A) /

280 Watts (20V/14A) / 250 Watts (20V/12.5A) by model.

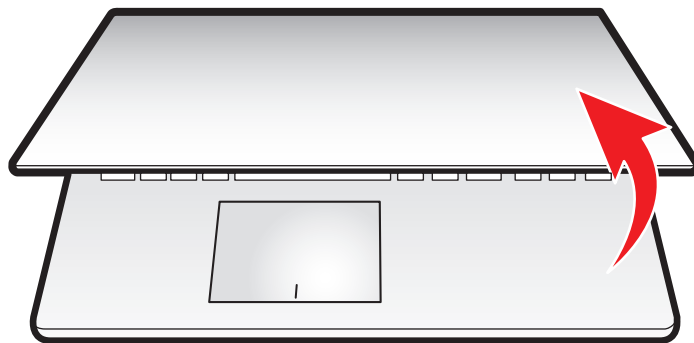
2.2 Preparing your Computer

- A** Connect the AC adapter's DC output plug to the DC IN jack.

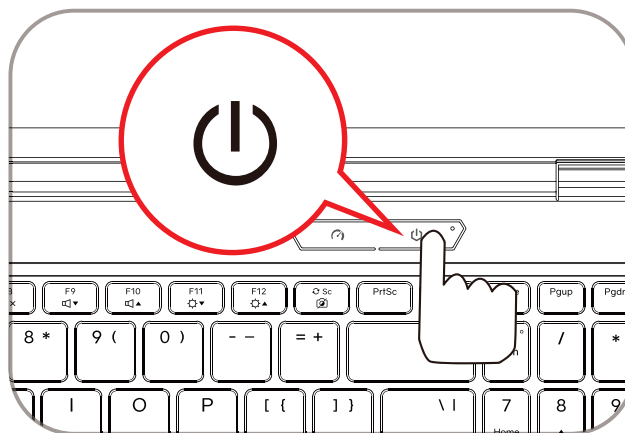
The first time to use, the computer will take a long time to boot up, please wait patiently.



- B** The display panel can be opened to a wide range of angles for optimal viewing.



- C** Press the power button to turn on your notebook computer.



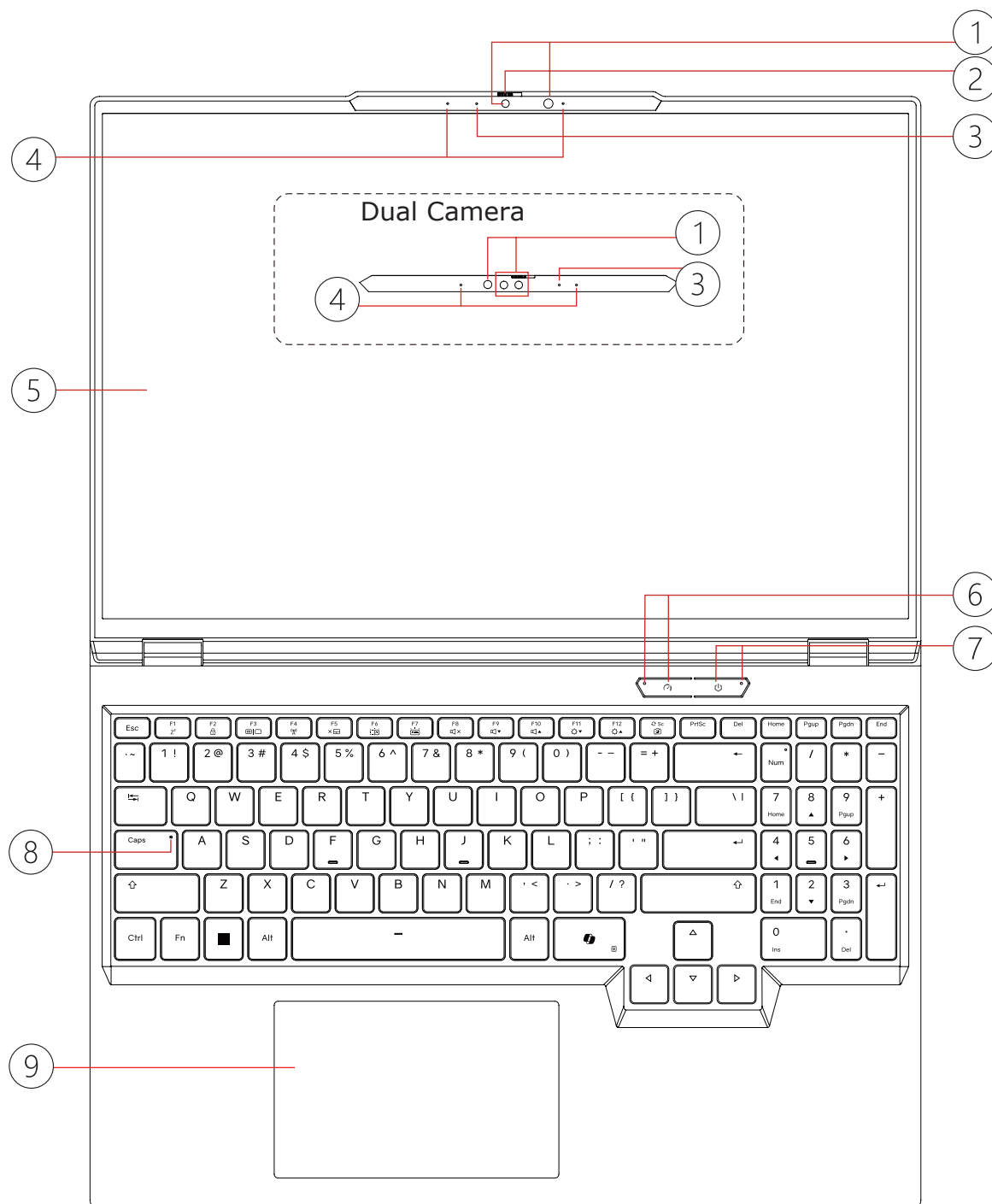
2.3 Product Overview

Please become familiar with each component before you operate the computer.

Top-Open View



NOTE: The product's color/ LED color, I/O port, indicator location, and specification will depend upon the actually shipping product.



① **Camera / Dual camera (option)**

A device that allows you to record video or take photographs with your computer.

Infrared capability for facial recognition camera (option)

can be used to identify and authenticate user to unlock notebook, conduct payment and other security functions.

② **Shutter latch**

Use Shutter latch to close the camera to protect privacy.

③ **Camera status LED**

The Camera Status LED shows the Camera status.

④ **Microphone**

Built-in microphones.

IR CCD MIC*2

⑤ **LCD screen**

Displays of your notebook computer.

⑥ **Custom / Turbo / Balance / Office mode switch**

Office mode: Green always on

Balance mode: Blue always on

Turbo mode: Purple always on

Custom mode: White always on

⑦ **Power button**

Press this button to turn the computer's power on or off.

Power indicator

No charging:

Power ON: Blue

Power Off: OFF

⑧ **Keyboard**

The keyboard provides keys with comfortable travel

(The keyboard legend will depend on the spec / region)

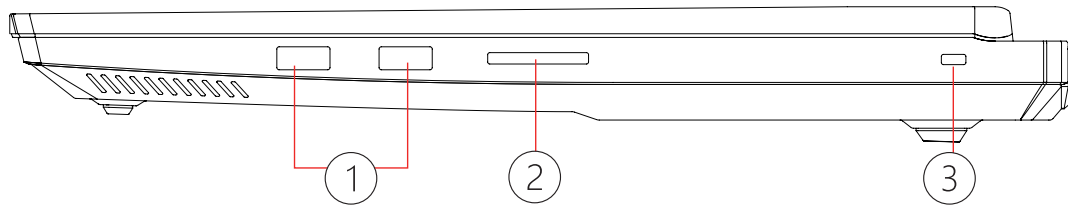
⑨ **Touchpad/Click pad**

Touch-sensitive pointing device which functions like the mouse.



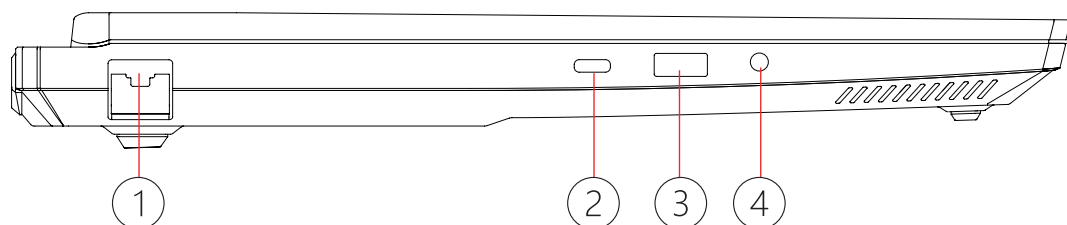
CAUTION:When you are not using the computer, keep the LCD screen closed to protect it from dust.

Right Side View



- ① USB port**
Connects a USB device (such as USB Zip drive, keyboard or mouse) into this jack.
With charging function.
- ② Card reader (option)**
 - USB interface, SD/SDHC/SDXC, SD7.0
 - Connector: Full SD size
- ③ Kensington® lock port**
To be secured using Kensington® Lock Port security products.

Left Side View



①

Network jack

This jack lets you connect to a LAN.

②

USB port (Type C)

Connects a USB device.

With PD 100W supported and charging function support 20V only.

③

USB port

Connects a USB device (such as USB Zip drive, keyboard or mouse) into this jack.

④

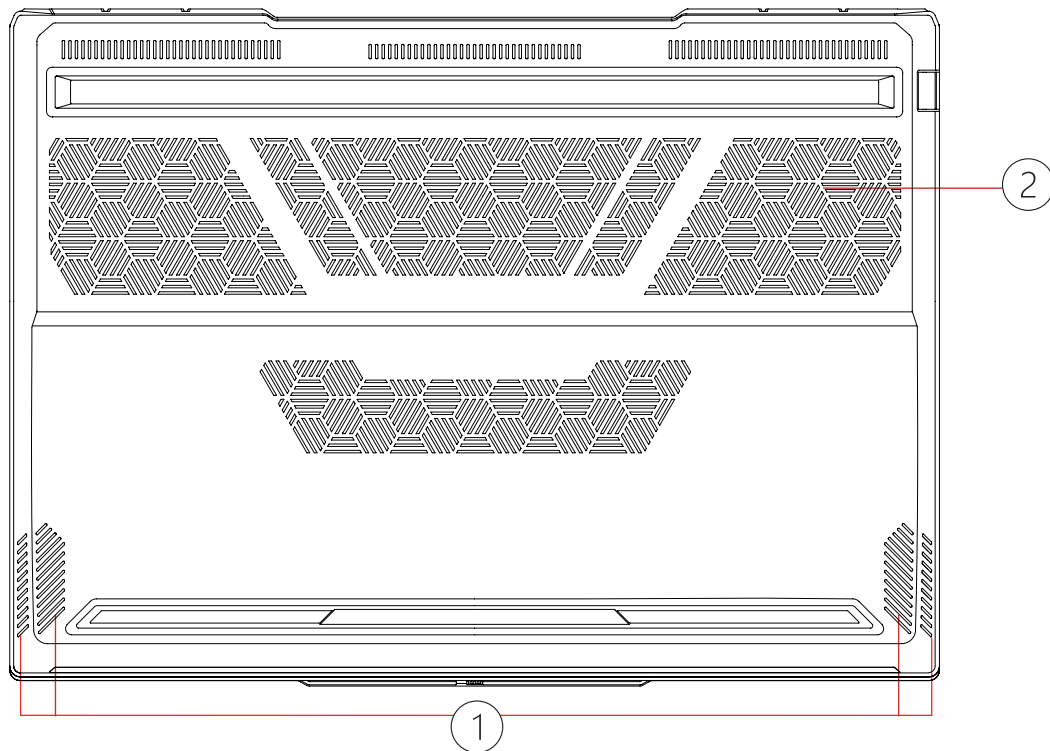
3.5mm UAJ jack

Connect the 3.5mm audio combo jack/headphones to this port.

Bottom Side View



NOTE : The product's thermal vent will depend upon the actually shipping product.



① **Speaker**

② **Vents**

The thermal vents are designed to cool the internal components and avoid overheating.

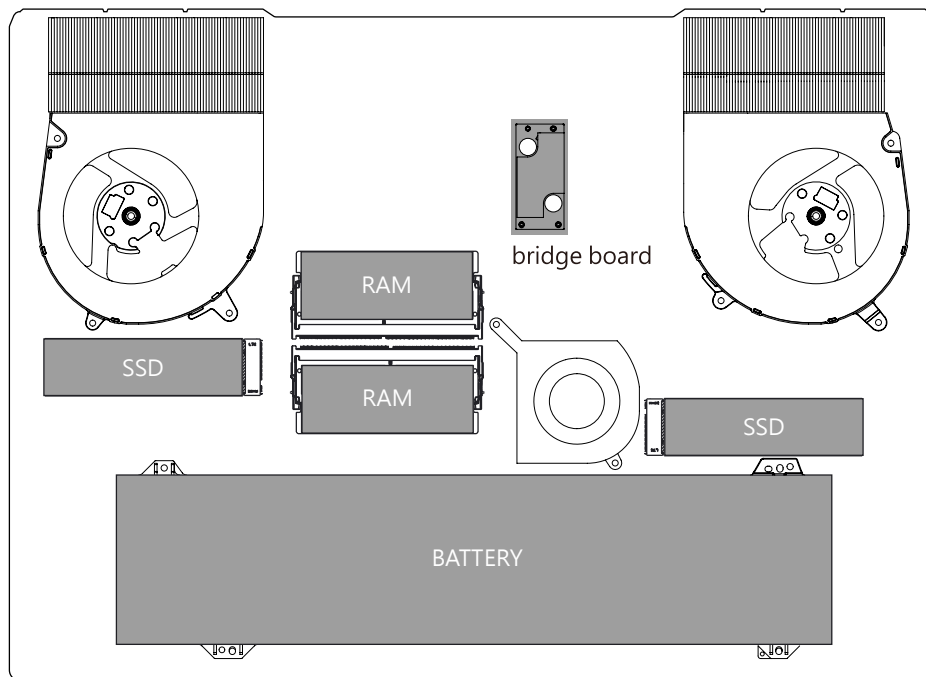


CAUTION: To reduce the possibility of heat-related injuries or of overheating the computer, do not place the computer directly on your lap or obstruct the computer air vents. Use the computer only on a hard, flat surface. Do not allow another hard surface, such as an adjoining optional printer, or a soft surface, such as pillows or rugs or clothing, to block airflow. Also, do not allow the AC adapter to come into contact with the skin or a soft surface, such as pillows or rugs or clothing, during operation.

Compartment View



NOTE: Specification will depend upon the actually shipping product.

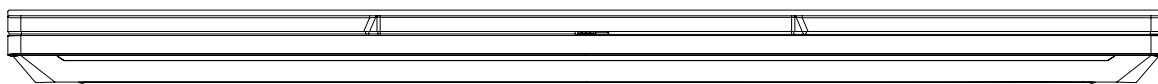


Please contact technical support team and after-sales service for any technical issues.

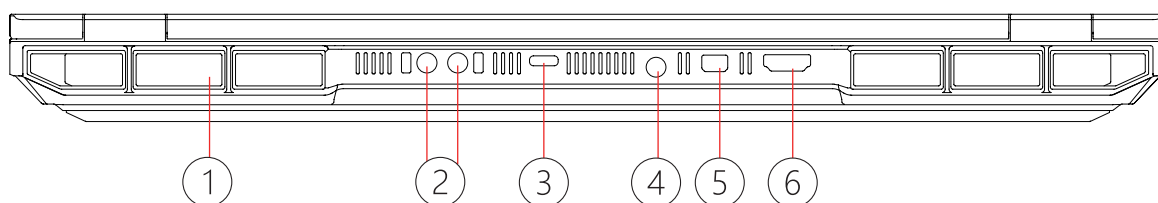


Do not dismantle the interior components (eg. Bridge board....etc) as this may result in severe consequences such as electrical short circuit and possibility of high temperature oxidation which will void the warranty.

Front Side View



Back Side View



① Vents

The thermal vents are designed to cool the internal components and avoid overheating.

② Cooling connector heads

If you have purchased a Liquid Cooling Box, please refer to the relevant manual.

③ **USB port (Type C)**

Connects a USB device.

PD140Wsupported(option) and With charging function support 20V only.

④ **Power connector**

Connects the AC adapter into this connector.

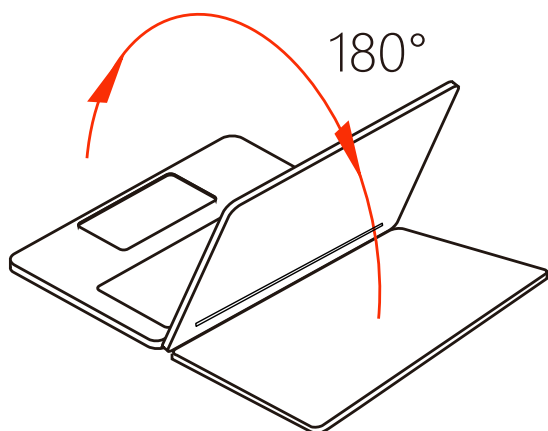
⑤ **Mini display port**

Supports high-definition digital video connections.

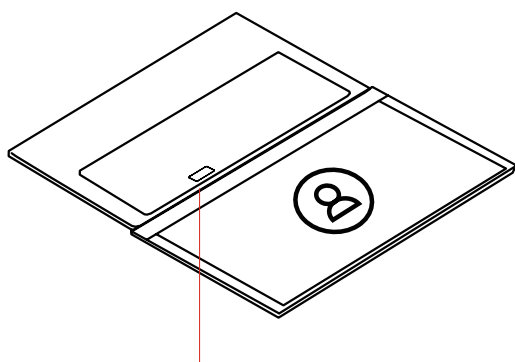
⑥ **HDMI port**

Connected to the screen, or home theater system.

Features



Rotate 180 degrees to share screen images.



Press button  Sc
can rotate the screen image

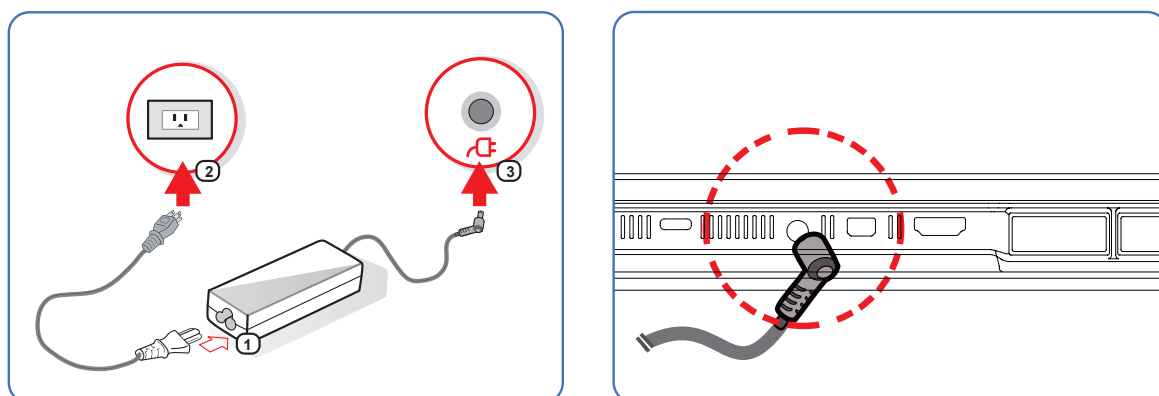
Chapter 3

Getting started

3.1 AC Adapter

Please be noted that it is strongly recommended to connect the AC adapter and use the AC power while using this notebook for the first time. When the AC adapter is connected, the battery is being charged immediately. Attach the AC adapter when you need to charge the battery or you want to operate from AC power. It is also the fastest way to get started, because the battery pack will need to be charged before you can operate from battery power.

Note that the AC adapter included in the package is approved for your notebook; using other adapter model may damage either the notebook or other devices attached to it.



CAUTION: The use of inferior extension cords may result in damage to your notebook. Your notebook comes with its own authorized AC adapter. Use of a different AC adapter or cable extension which is not authorized for use will void warranty protection if damage to hardware is found in association to said adapter or extension cable.

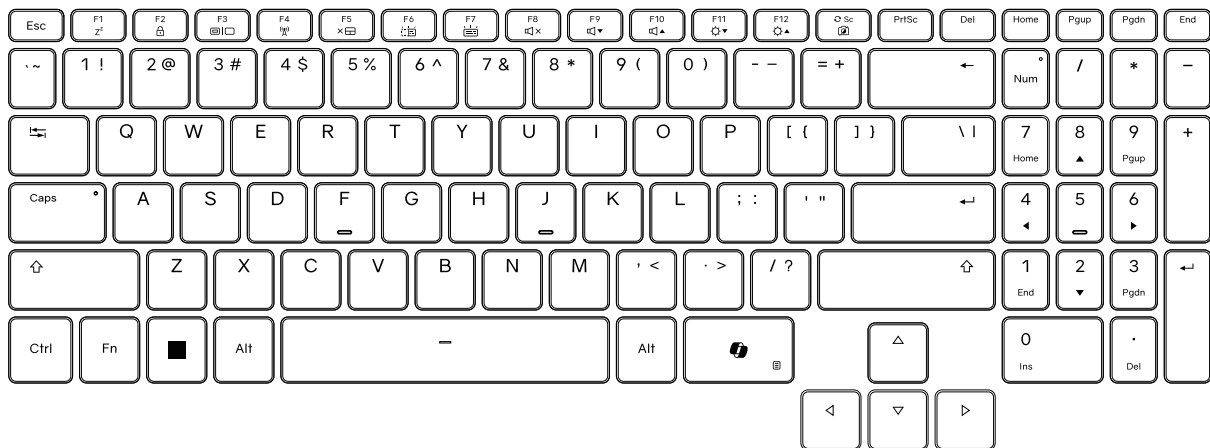


NOTE: The power adapter can become hot when in use. Please be sure the AC adapter is not covered with any materials keep it away from exposed parts of your body. The AC adapter appearance may vary depending on your region.

3.2 Knowing the Keyboard

The following defines the hot keys on the Keyboard.

The commands can only be accessed by first pressing and holding the function key while pressing a key with a command.

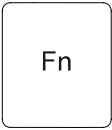
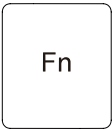
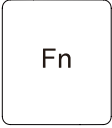
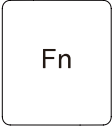


NOTE: The design of the number of keys available on the keyboard, printing color or backlight depends on the country (region) in which the keyboard was designed and built.

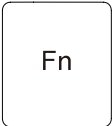
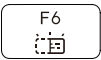
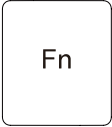
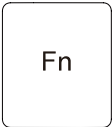
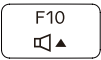
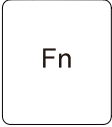
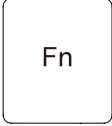
3.2.1 For keyboard users

To activate these functions, press the hot key associated with the desired function as below :

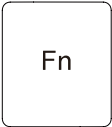
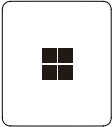
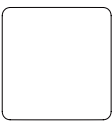
Function Keys

Keypad	Function Description
 +  Suspend:	Press this key combination (Fn+F1) to enter sleep mode.
 +  Windows Lock/On:	Press this key combination (Fn+F2) to turn Windows Lock/On .
 +  Display mode:	Press this key combination (Fn+F3) to enable Display Mode.
 +  RADIO On/Off:	Press this key combination (Fn+F4) to turn all radios on or off.
 +  Touch pad On/Off:	Press this key combination (Fn+F5) to On/Off Touch pad mode.

Function Keys

Keypad	Function Description
 + 	Keyboard Brightness down:
 + 	Keyboard Brightness up:
 + 	MUTE:
 + 	Volume down:
 + 	Volume up:
 + 	Brightness down:
 + 	Brightness up:

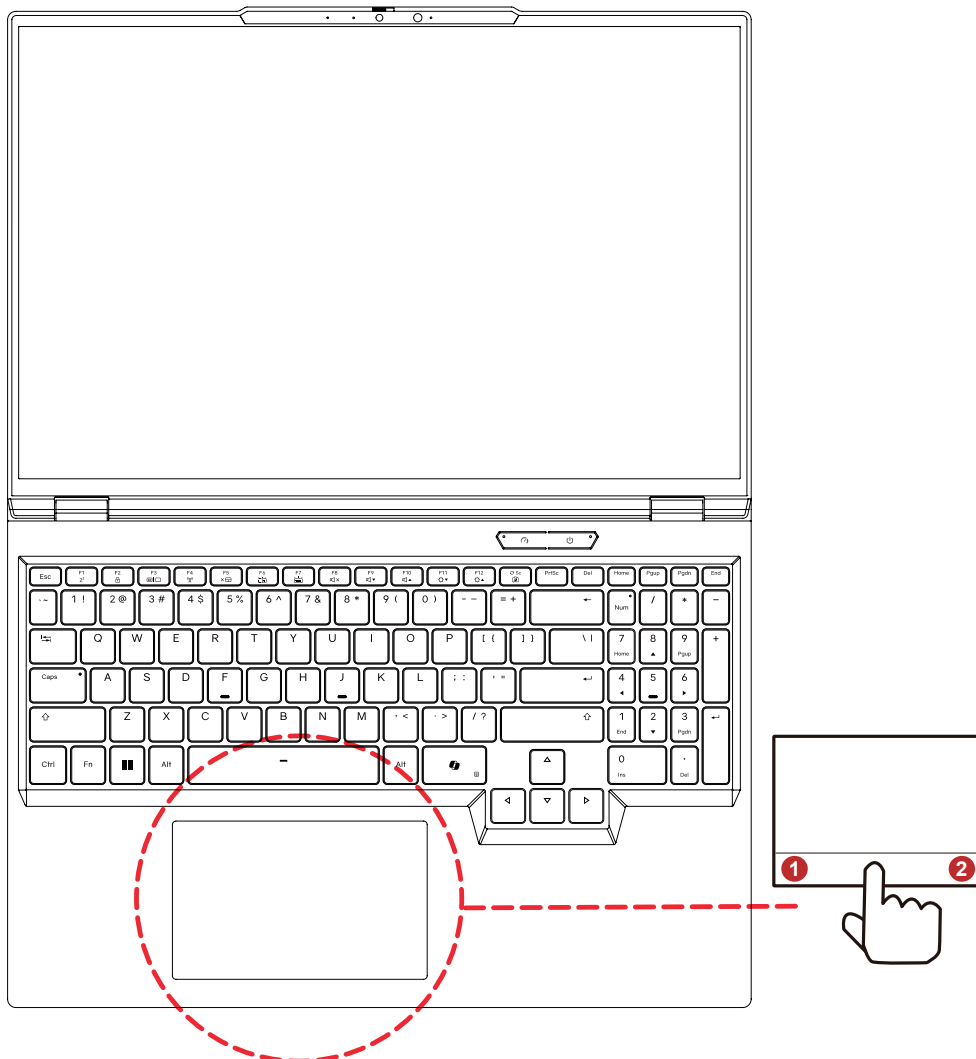
Function Keys

Keypad	Function Description
   Camera On/Off:	Press this key to On/Off camera function.
   Search Charm : (Option)	For a full list of Windows keyboard shortcuts, please visit the Microsoft official website https://support.microsoft.com/en-us/
	Enable Copilot on Windows Copilot will be displayed on the right side of the computer, Copilot supports pictures , voice, text and other chat methods.

3.3 Using the touchpad / clickpad

The touchpad / clickpad is a rectangular electronic panel located just below your keyboard. You can use the buttons below the touchpad as left and right mouse buttons.

Press the left **1** and right **2** buttons located on the edge of the touchpad / clickpad to make selections and run functions. These two buttons are similar to the left and right buttons on a mouse. Tapping on the touchpad / clickpad produces similar results.



Chapter 4

BIOS setup

4.1 About BIOS Setup

4.1.1 When to Use BIOS Setup ?

You may need to run the BIOS Setup when:

- An error message appears on the screen during the system booting up and is requested to run SETUP.
- You want to change the default settings for customized features.
- You want to reload the default BIOS settings.

4.1.2 How to Run BIOS Setup ?

To run the BIOS Setup Utility, turn on the notebook and press the [Del] key during the POST procedure.

If the message disappears before you respond and you still wish to enter Setup, either restart the system by turning it OFF and ON, or simultaneously pressing [Ctrl]+[Alt]+[Del] keys to restart.

Be noted that the screen snaps and setting options in this chapter are for your references only. The actual setting screens and options on your Notebook may be different because of BIOS update.

The setup function only can be invoked by pressing [Del] or [F2] key during POST that provide a approach to change some setting and configuration the user prefer, and the changed values will save in the NVRAM and will take effect after the system rebooted. The setup uses a menu interface to allow the user to configure their system and the features are briefly listed as follow.

Press [F7] key for Boot Menu.



NOTE : The drivers, BIOS and utilities bundled in the support DVD may vary by models and are subject to change without notice

4.2 BIOS Setup Menu

Once you enter the BIOS Setup Utility, the Main Menu will appear on the screen. Select the tags to enter the other menus.

Main Menu

Show system overview about memory size and setting of system time and date.

Advanced Menu

Display device information and advanced setting items.

Security Menu

Install or clear the password settings for supervisor and user.

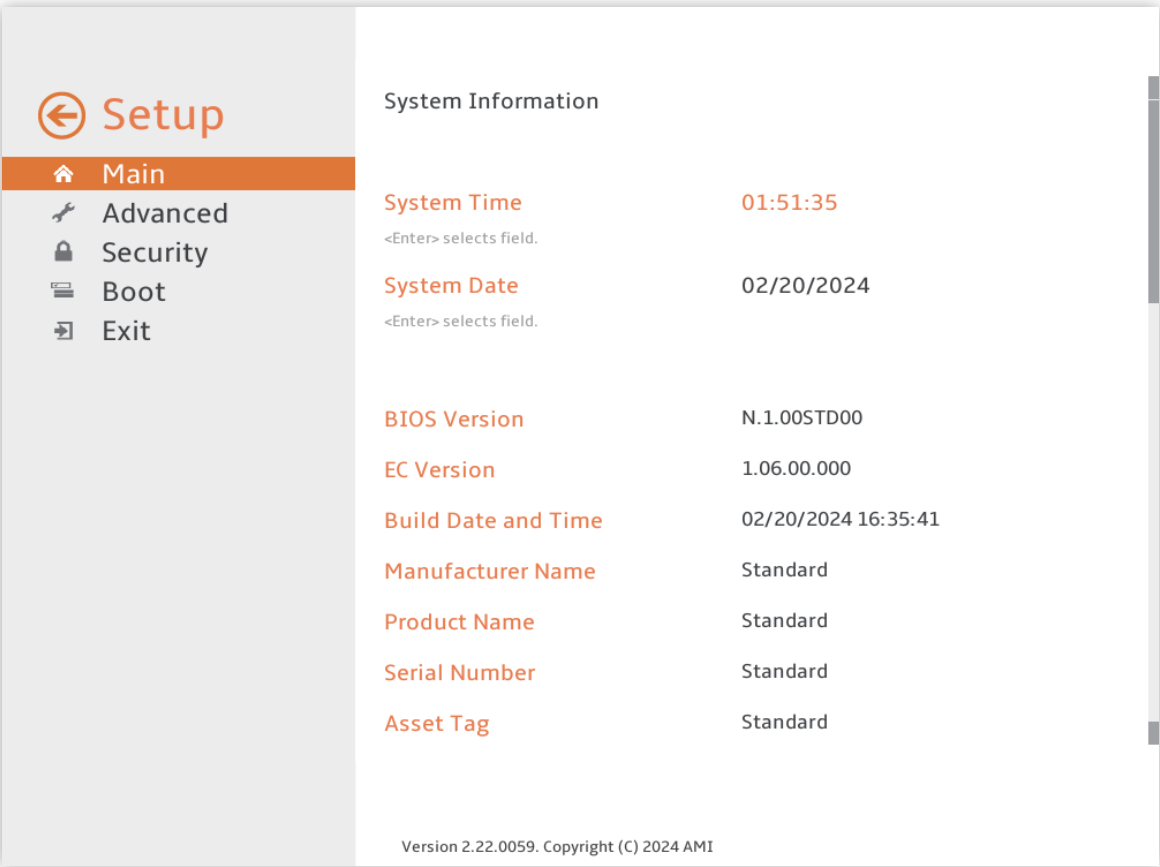
Boot Menu

Configure Settings during System Boot.

Exit Menu

Save or discard the changes before leaving the BIOS Setup Menu.

4.2.1 Main Menu



- System Time

This item allows you to set the system time. There is a small internal (CMOS) battery which is designed to maintain your system clock. It is designed to maintain time even when the PC is powered down or in sleep mode. The time format is [hour:minute:second].

Use [+] or [-] to configure system Time.

- System Date

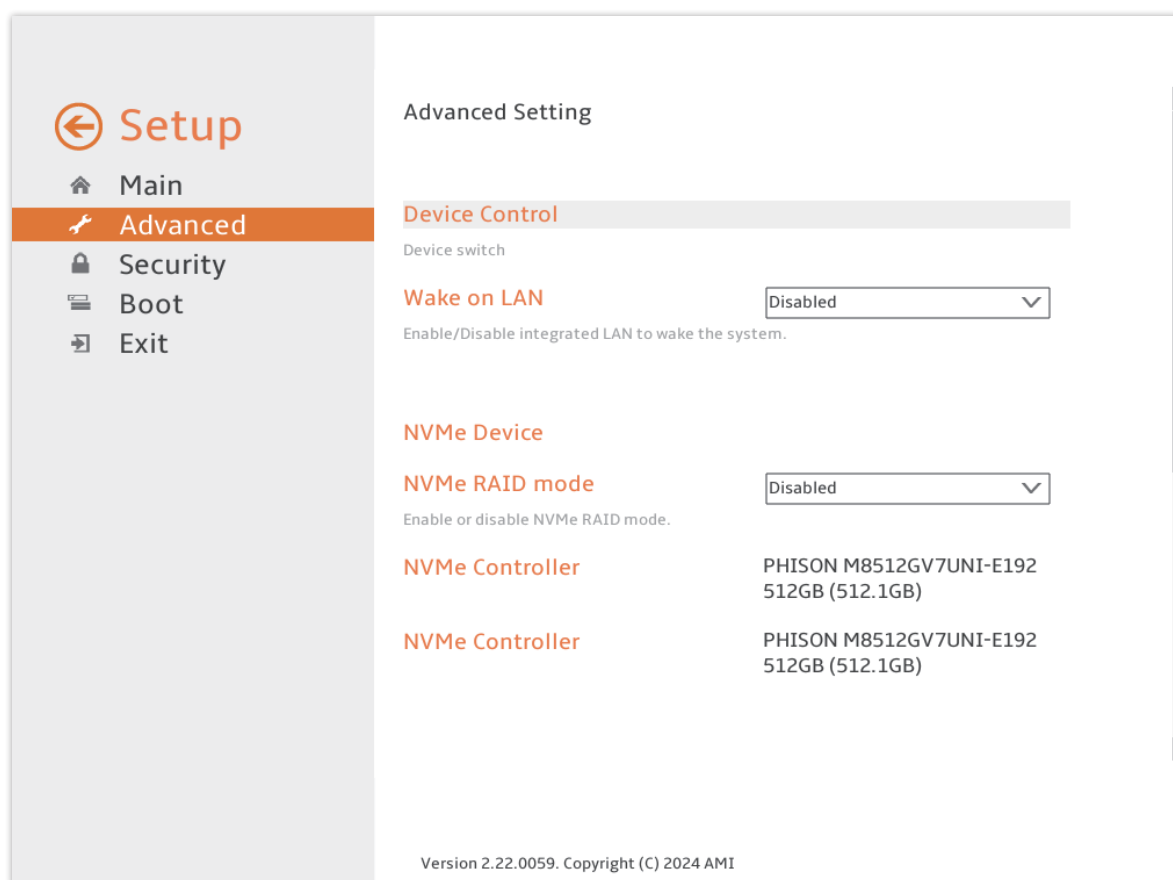
This item allows you to set the system date. The date format is [day:month:date:year].

Use [ENTER], [TAB] or [SHIFT-TAB] to select a field.

Month (Month)	The month from 01 (January) to 12 (December).
Date (Date)	The date from 01 to 31.
Year (Year)	The year can be adjusted by users.

- Processor Type: { Brand information of installed processor }
- Processor Speed: { Current Processor Speed, unit is MHz }
- Total Memory: { This allows you to see the total amount of memory }
- Memory Frequency: { Current Memory Speed, unit is MT/s }

4.2.2 Advanced Menu



- Device Control
Onboard Device Component(s) Switch and may include the following: Webcam/Wi-Fi/Bluetooth/HD Audio/Onboard LAN/Card Reader/USB.....
- Wake on LAN
Enable/Disable integrated LAN to wake system.
- NVME Device
NVME SSD info will be displayed
- NVME RAID mode
Enable or disable NVMe RAID mode
- Operating mode
Enable Office Mode/Balance Mode/Turbo Mode
Operating Mode includes the Fan Speed and CPU power level setting to optimize overall performance and heat dissipation

4.2.3 Security Menu

← Setup

Home Main

Wrench Advanced

Security

Boot

Exit

Security Settings

Access Level

Administrator

Administrator Password

Not Installed

User Password

Not Installed

Change Administrator Password -Not Installed

Set Setup Administrator Password

Change User Password -Not Installed

Set User Password

Password Login Control

Both

Enables password entry on boot, setup page, or both.

SCPC Firmware Protection Support

Enabled

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← Setup

Home Main

Wrench Advanced

Security

Boot

Exit

Vendor Keys

Valid

Factory Key Provision

Enabled

Install factory default Secure Boot keys after the platform reset and while the System is in Setup mode

Restore Factory Keys

Force System to User Mode.
Install factory default Secure Boot key databases

Reset To Setup Mode

Delete all Secure Boot key databases from NVRAM

Secure Boot variable

| Size| Keys| Key Source

Platform Key

(PK)| 881| 1| Factory

Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate:
a)EFI_SIGNATURE_LIST
b)EFI_CERT_X509 (DER)
c)EFI_CERT_RSA2048 (bin)
d)EFI_CERT_SHAXXX

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- Change Administrator Password

When this item is selected, a message box shall appear on the screen as below:

Enter New Password

Type a maximum of 20-digit password and press [Enter].

The password typed now will replace any previously set password from CMOS memory. You may also press [ESC] to abandon new password setting. It is extremely important to keep record for any BIOS password which is set as it can not be reset even with the removal of a CMOS battery for the security of your PC.

Select Change User Password to give or to abandon password setting same as Change Administrator Password item above.

Note that Administrator Password field allows users to enter and change the settings of the BIOS SETUP UTILITY, while User Password field only allows users to enter the BIOS SETUP UTILITY without having the authorization to make any change.

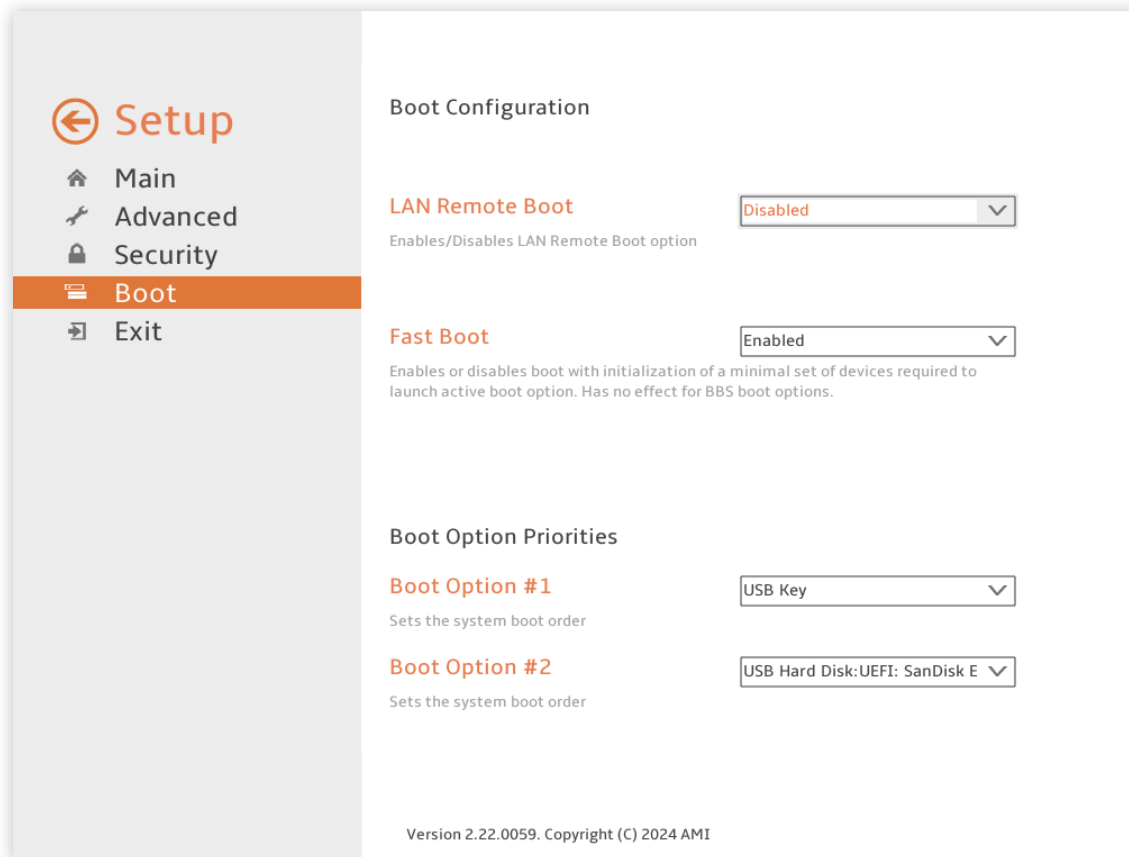
The Password Check item is used to specify the type of BIOS password protection that is implemented.

To clear a set Administrator Password/ User Password, just press [Enter] under Change Administrator Password/ Change User Password field when you are prompted to enter the password. A message box will pop up confirming password will be disabled. Once the password is disabled, the system will boot and user can enter setup without entering password.

Key Management:

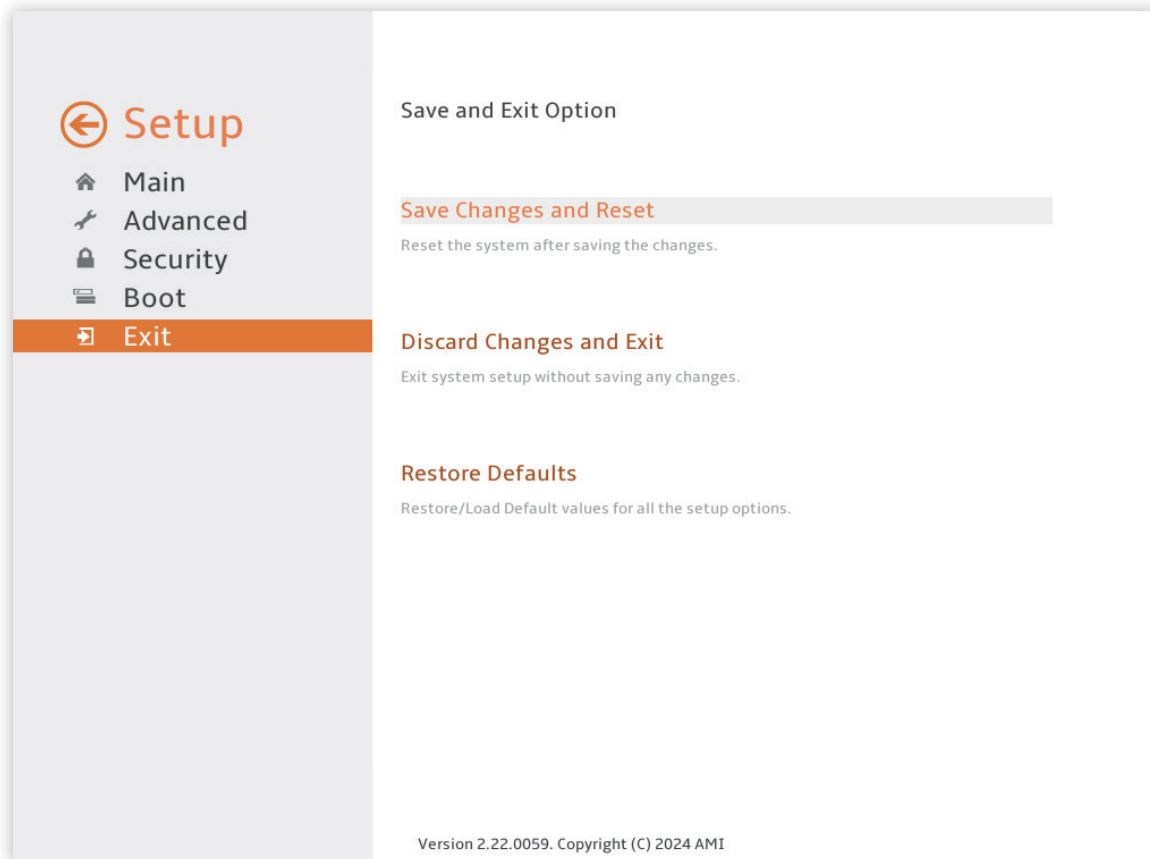
Enables expert user to modify Secure Boot Policy variables without full authentication.

4.2.4 Boot Menu



- Boot Configuration
Configure Settings during System Boot.
- Set Boot Priority (1st/2nd/3rd/..... Boot)
Specifies the boot sequence from the available devices.
A device enclosed in parenthesis has been disabled in the corresponding type menu.
- Hard Disk Drive / USB HardDisk Drive
Specifies the Boot Device Priority sequence.
- Fast Boot: [Enabled/ Disabled]

4.2.5 Exit Menu



- **Save Changes and Reset**
Exit system setup after saving the changes. F10 key can be used for this operation.
- **Discard Changes and Exit**
Exit system setup without saving any changes.
ESC key can be used for this operation.
- **Restore Defaults**
Restore/Load Defaults values for all the setup options. F9 key can be used for this operation.